

A Correlation for Laminar Hydrodynamic Entry Length Solutions for Circular and Noncircular Ducts

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Laminar hydrodynamic entry length solutions for circular and noncircular ducts are essential in proper design of compact heat exchangers and other heat transfer and fluid flow devices. A closed form equation has been proposed to present these solutions for the circular tube, parallel plates, rectangular, equilateral triangular, and concentric annular ducts. The necessary constants are evaluated and it is shown that the proposed correlation predicts the apparent friction factors within ± 2.4 percent.

Introduction

In the current energy conscientious era, the use of compact heat exchangers is increasing at an accelerated pace due to their relatively lighter weight and lower cost. The design Reynolds number range for these exchangers usually falls well within the laminar flow regime because of the associated small flow passage hydraulic radius and gas flows. Most of the compact heat exchangers employ heat transfer surfaces which are interrupted in the flow direction such as louvered and strip fin surfaces. Usually only the highly compact heat exchangers have the continuous cylindrical passages of different shapes. For the interrupted surfaces, a new hydrodynamic (and also thermal) boundary layer develops on the surface after each interruption. Hence the developing laminar internal flow solutions are useful in the design of compact heat exchangers. Laminar hydrodynamic entry length solutions¹ are available in the literature for a circular tube, parallel plates, and rectangular, triangular, concentric annular and eccentric annular ducts. These solutions have been compared and critically reviewed by Shah and London [1].²

The objective of this paper is to provide a correlation to determine the friction factors (or pressure drop) in the hydrodynamic entry length region for laminar flow through circular and non-circular ducts. In all of the solutions used for the correlation,

it is idealized that the fluid is Newtonian and incompressible, fluid properties are constant, and body forces are neglected. The velocity profile at the entrance of the duct ($x = 0$) is considered uniform.

Dimensionless Pressure Drop

A fluid mechanics or heat transfer equipment designer is interested in determining the pressure drop in the entrance region for a given application. This pressure drop is presented by at least three alternate dimensionless groups: Δp^* , $f_{app} Re$ or $K(x)$. These are related to each other as follows.

$$\Delta p^* = \frac{\Delta p}{(\rho u_m^2 / 2g_c)} = (f_{app} Re)(4x^+), \quad (1)$$

and

$$\Delta p^* = (f Re)(4x^+) + K(x), \quad (2)$$

where f_{app} is an apparent Fanning friction factor and is based on the total pressure drop from $x = 0$ to x . The f_{app} takes into account both the skin friction and the change in momentum rate in the hydrodynamic entrance region. f stands for the Fanning friction factor for the fully developed region. Incremental pressure drop number $K(x)$ represents an additional dimensionless pressure drop due to momentum change and accumulated increment in wall shear between a developing and a fully developed flow. The value of $K(x)$ increases monotonically from 0 at $x = 0$ to a constant value (designated as $K(\infty)$) in the fully developed region. The pressure drop for a long duct ($L > L_{hy}$)³ can be ac-

¹The hydrodynamic entry length region of a duct is that region where the velocity boundary layer is developing, from zero thickness at the entrance to the thickness equal to the pipe radius far downstream. In this region, the fluid velocity profile changes from the initial profile at the entrance to an invariant form downstream.

²Numbers in brackets designate References at end of paper.

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³The hydrodynamic entry length L_{hy} may be defined somewhat arbitrarily as the duct length required to achieve a duct cross section maximum velocity of 99 percent of the corresponding fully developed magnitude, when the entering flow is uniform.

curately calculated from equation (2) with the knowledge of f Re and $K(\infty)$.

Hydrodynamic Entry Length Solutions for the Circular Tube

The development of the proposed correlation is based on the following results for the circular tube.

Shapiro, et al. [2] carried out a careful experimental investigation of the hydrodynamic entrance region of a circular tube. They correlated the experimental data as follows for $10^{-5} \leq x^+ \leq 10^{-3}$.

$$\Delta p^* = 13.74\sqrt{x^+}, \quad (3)$$

or

$$f_{app} = \frac{3.44}{\sqrt{Re_x}} = \frac{3.44}{Re\sqrt{x^+}}, \quad (4)$$

where the range of Re was from 51,200 to 113,400. Even at this high Reynolds number, for which the fully developed flow is turbulent, a laminar boundary layer is formed at the entry of a tube. This layer increases in thickness along the length of the tube and eventually at $Re_x \simeq 5 \times 10^5$ (depending on the entrance condition of the fluid and tube) undergoes transition and becomes turbulent.

The mean skin friction factor for laminar flow over a flat plate with zero pressure gradient is given by [3]

$$f_m = \frac{1}{x} \int_0^x f_x dx = \frac{1.328}{\sqrt{Re_x}}. \quad (5)$$

A comparison of equations (4) and (5) reveals that the apparent friction factor for a pipe entry is 2.59 times as great as the mean skin friction factor for a flat plate. Shapiro, et al. [2] showed that when the boundary layer is thin compared to the pipe radius, the boundary layer behavior in a tube entry is substantially identical with that on a flat plate. Consequently, the skin friction coefficients due to wall shear stress are the same for the tube entry and flat plate. In the tube entrance with $x^+ < 10^{-3}$, 39 percent (1.328/3.44) of the pressure drop is caused by wall friction and the remaining 61 percent by an increase in momentum flux. The fractional contribution of the momentum flux change to the total pressure drop decreases with increasing x^+ and approaches zero in the fully developed region.

Hornbeck [4], and Shah and Farnia [5] solved the nonlinear momentum equation numerically by finite difference methods. The f_{app} Re factors of Shah and Farnia are tabulated by Liu [6] and Shah and London [1]. Hornbeck's f_{app} Re factors are about

4.6 percent higher than those of equation (4) for $x^+ \leq 0.001$. Liu's f_{app} Re factors are lower than those of equation (4), ranging from 2.4 percent at $x^+ \simeq 0.0002$ to 0.3 percent at $x^+ = 0.001$. Thus an excellent agreement has been achieved between the numerical and experimental f_{app} Re factors. The hydrodynamic entrance length $L_{hy}^+ (= L_{hy}/D_h \text{ Re})$ calculated from the results of Hornbeck and Liu are 0.056 and 0.054, respectively.

From the above discussion, Δp^* in the entrance region of a circular tube can be accurately determined from equation (3) for $x^+ \leq 10^{-3}$. For a "long duct" with $x^+ \geq 0.06$, Δp^* can be determined by equation (2) for known values of f Re ($=16$) and $K(\infty)$. Many researchers, while theoretically investigating the entry flow problem, have determined the values of $K(\infty)$ varying from 1.08 to 1.41; while the mean experimental value varies from 1.20 to 1.32 [7]. Bender [8] theoretically obtained $K(\infty) = 1.25$ which also approximates the mean of experimental values. Thus, based on this last value, for $x^+ \geq 0.06$,

$$\Delta p^* = 1.25 + 64x^+. \quad (6)$$

Bender then combined equations (3) and (6) as follows to cover the complete range of x^+ .

$$\Delta p^* = 13.74\sqrt{x^+} + \frac{1.25 + 64x^+ - 13.74\sqrt{x^+}}{1 + 0.00018(x^+)^2}. \quad (7)$$

The present author modified the constant 0.00018 to 0.00021 so that the calculated Δp^* provided a least r.m.s. error when compared with Δp^* of Liu [6]. The f_{app} Re factors based on this modified equation agree with those of Liu within ± 1.9 percent for $x^+ > 0.00022$.

The Proposed Correlation and Results

At the entrance of the duct, the velocity boundary layer starts developing at each wall under the imposed flow acceleration. As long as the thickness of the boundary layer is small compared to the duct dimensions, the boundary layers from different walls do not affect each other appreciably. As mentioned earlier, Shapiro, et al. [2] state that the boundary layer behavior in a tube entry is substantially identical with that on a flat plate (external flow), and the tube curvature has negligible effect for $x^+ \leq 0.001$. For a "long" duct, Δp^* can be accurately calculated from equation (2) with $K(x)$ replaced by $K(\infty)$. Hence, extending Bender's idea, the following simplified equation is proposed for predicting f_{app} Re factors in the hydrodynamic entry region of the circular and noncircular ducts.

$$f_{app} \text{ Re} = \frac{3.44}{\sqrt{x^+}} + \frac{(f \text{ Re}) + K(\infty)/(4x^+) - 3.44/\sqrt{x^+}}{1 + C(x^+)^2}. \quad (8)$$

Nomenclature

C = a constant used in equation (8), dimensionless
 D_h = hydraulic diameter of the duct passage, m
 f = $\tau/(\rho u_m^2/2g_c)$ = "Fanning" or "small" friction factor, for fully developed flow if no subscript used; f_x represents the local friction factor in the hydrodynamic entrance region, dimensionless
 f_{app} = apparent Fanning friction factor, defined by equation (1), dimensionless
 $K(x)$ = incremental pressure drop num-

ber, defined by equation (2), dimensionless
 L = length of the duct, m
 Δp = fluid static pressure drop in the flow direction from the entrance to the cross section of interest, Pa
 $\Delta p^* = \Delta p/(\rho u_m^2/2g_c)$, dimensionless
 $Re = \rho u_m D_h/\mu$ = Reynolds number, $Re_x = \rho u_m x/\mu$, dimensionless
 r^* = the ratio of inner to outer radius of the concentric annular duct, dimensionless
 u_m = fluid mean axial velocity in the duct, m/s

x = axial (flow direction) coordinate in Cartesian or cylindrical coordinate systems, m
 $x^+ = x/(D_h \text{ Re})$, dimensionless
 α^* = aspect ratio, the ratio of the short side to long side of the rectangular duct, dimensionless
 μ = fluid dynamic viscosity coefficient, Pa·s
 ρ = fluid density, kg/m³
 τ = wall shear stress due to skin friction, Pa
 ϕ = half apex angle of the isosceles triangular duct, deg

Table 1 $K(\infty)$, f Re, and C to be used in equation (8) for rectangular, triangular, and concentric annular ducts

	$K(\infty)$	f Re	C	% error
Rectangular ducts				
α^*				
1.00	1.43	14.227	0.00029	± 2.3
0.50	1.28	15.548	0.00021	± 1.9
0.20	0.931	19.071	0.000076	± 1.7
0.00	0.674	24.000	0.000029	± 2.4
Equilateral triangular duct				
2ϕ				
60°	1.69	13.333	0.00053	± 2.4
Concentric annular ducts				
r^*				
0	1.25	16.000	0.000212	± 1.9
0.05	0.830	21.567	0.000050	± 2.0
0.10	0.784	22.343	0.000043	± 1.9
0.50	0.688	23.813	0.000032	± 2.2
0.75	0.678	22.967	0.000030	± 2.1
1.00	0.674	24.000	0.000029	± 2.4

Here $K(\infty)$ and f Re are for hydrodynamically fully developed flow; C is a constant dependent upon the duct geometry and is generally obtained by comparing the above approximate f_{app} Re with that of the analytical entry length solution. The values of the coefficient C are provided below for parallel plates, rectangular, triangular, and concentric annular ducts. f Re, $K(\infty)$, and C are summarized in Table 1 for a quick reference.

Parallel Plates. The most accurate hydrodynamic entry length solutions, based on the boundary layer type idealizations, are due to Bodoia and Osterle [9], and Liu [6]. They obtained the solution by finite difference methods. The f_{app} Re factors of Bodoia and Osterle and of Liu are reported by Shah and London [1]. These factors of Bodoia and Osterle and of Liu are in excellent agreement. When compared to equation (4), the f_{app} Re factors of Liu vary from 1 percent higher at $x^+ = 0.0002$ to 1 percent lower at $x^+ = 0.001$. The hydrodynamic entrance length L_{hy} of [6, 9] is 0.011.

For a long duct with $x^+ \geq 0.01$, f_{app} Re can be determined accurately by equation (2) for known values of f Re ($=24$) and $K(\infty)$. The theoretically determined values of $K(\infty)$ generally vary from 0.60 to 0.74 [7]. As no experimental value of $K(\infty)$ exists, the mean of theoretical values, 0.66, is recommended for $K(\infty)$. Using these values of f Re and $K(\infty)$, the constant C of equation (8) was found to be 0.000029 by a least square fit to f_{app} Re of Liu. With this value of C , the f_{app} Re of equation (8) agree with those of Liu within ± 2.4 percent (within ± 1.5 percent for all but three points).

Noncircular Ducts. The hydrodynamic entry length solutions for rectangular, equilateral or isosceles triangular and concentric annular ducts have been obtained by numerous investigators. These solutions are summarized and compared by Shah and London [1].

For rectangular ducts, the theoretical results of Curr, et al. [10] and experimental results of Beavers, et al. [11] are believed to be the most accurate; the agreement between these results is excellent. The f_{app} Re factors, based on the graphical Δp^* of Curr, et al., are in excellent agreement with those of equation (4) for $x^+ \lesssim 0.003$.

For an equilateral triangular duct, the theoretical f_{app} Re factors of Aggarwala and Gangal [12] are believed to be the most accurate and are reported in [1] for $x^+ \geq 0.015$. These data were joined asymptotically with equation (4) to arrive at the interpolated values of f_{app} Re factors for $x^+ \leq 0.010$.

For concentric annular ducts, the theoretical f_{app} Re factors of Liu [6] are the most accurate. When compared with equation (4), these factors vary from 1 percent higher at $x^+ = 0.0002$ to 1 percent lower at $x^+ = 0.001$.

With the known values of f Re and $K(\infty)$ as reported in Table 1, the constant C of equation (8) was determined for the above duct geometries by a least square fit to the f_{app} Re factors of the following investigators: Curr, et al. [10] for rectangular ducts, Aggarwala and Gangal [12] for the equilateral triangular duct, and Liu [6] for concentric annular ducts. The values of C thus obtained are presented in Table 1. The subsequent error between the predicted f_{app} Re factors by equation (8) and the corresponding theoretical values of the above investigators are also reported in Table 1.

Summary and Conclusions

The laminar hydrodynamic entry length solutions for the circular tube, parallel plates, rectangular, equilateral triangular, and concentric annular ducts can be presented in a closed form by equation (8), where the values of f Re, $K(\infty)$ and C are summarized in Table 1. The predicted f_{app} Re factors by equation (8) agree within ± 2.4 percent with the most accurate analytical results reported in the literature for the foregoing duct geometries. These f_{app} Re factors include the contributions of the skin friction and the change in momentum rate. The influence of form drag is not included in the correlation.

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⁴Increasing significant digits for the constant C does not produce improved accuracy in predicting f_{app} Re factors. Hence only up to six digits are provided for C in Table 1.